



Programming with Python

23. Schleifen mit for

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Dies ist ein Kurs über das Programmieren mit der Programmiersprache Python an der Universität Hefei (合肥大学).

Die Webseite mit dem Lehrmaterial dieses Kurses ist <https://thomasweise.github.io/programmingWithPython> (siehe auch den QR-Code unten rechts). Dort können Sie das Kursbuch (in Englisch) und diese Slides finden. Das Repository mit den Beispielprogrammen in Python finden Sie unter <https://github.com/thomasWeise/programmingWithPythonCode>.



Outline



1. Einleitung
2. Die for-Schleife
3. continue und break
4. Schleifen verschachteln
5. Iterieren über Sequenzen
6. Zusammenfassung





Einleitung



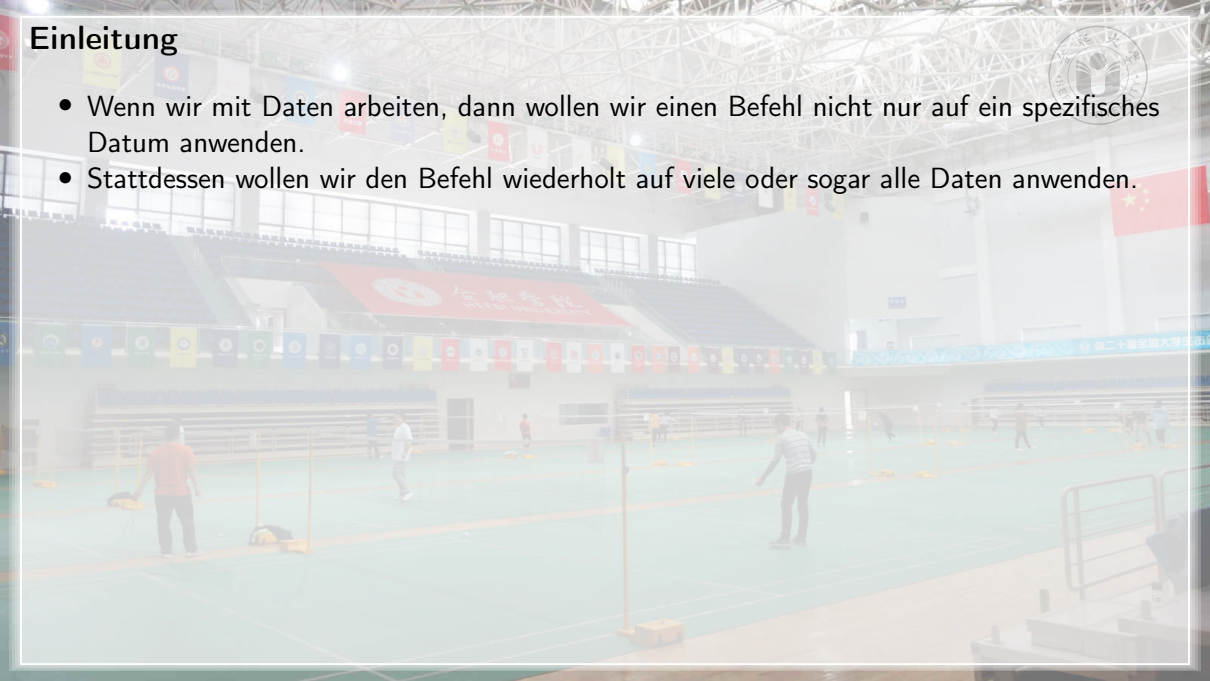
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Das *Structured Program Theorem* sagt aus, dass jede berechenbare Funktion mit nur drei verschiedenen Kontrollflusselementen berechnet werden kann, nämlich mit (1) dem sequenziellen Ausführen von Befehlen, (2) dem selektiven Ausführen von Befehlen (also mit Alternativen), und (3) dem iterativen (repetitiven) ausführen von Befehlen (bis eine bestimmte Bedingung eintritt)^{7,8,33,62}.

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- Danach sind wir im Prinzip in der Lage, *jede* Berechnung durchzuführen, die mit unseren Computern möglich ist.
- Alles was dann folgt sind nur Konzepte, die dafür da sind, unser Leben als Programmierer leichter zu machen.

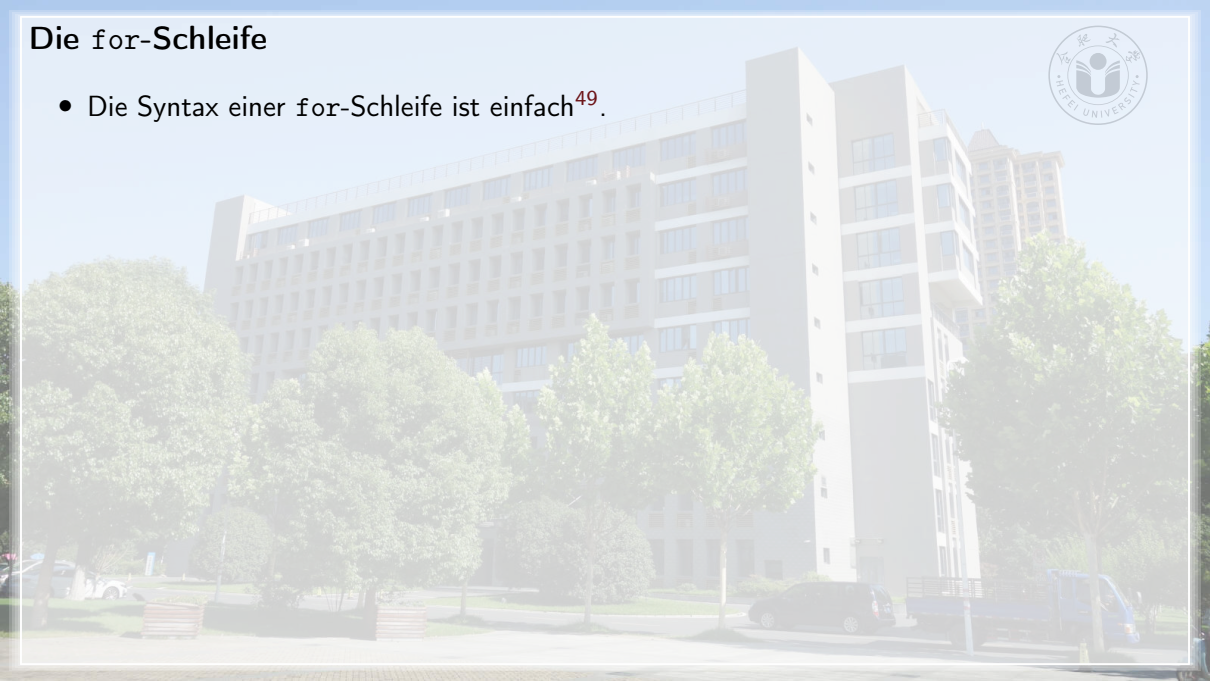


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- Nach dem Schleifenkörper folgt eine Leerzeile, nach der es dann mit normalem Code weitergeht.

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- Ranges, wie Slices, können auch mit negativen Schritten arbeiten: `range(40, 30, -3)` beginnt mit 40, steigt in Schritten von 3 ab, und hört vor 30 auf, ist also (40, 37, 34, 31).

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7      squares[i] = i * i # Stores 0: 0, 1: 1, 2: 4, 3: 9, 4: 16.
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9  for i in range(6, 9): # i takes on the values 6, 7, and 8 one by one.
10     squares[i] = i * i # Stores 6: 36, 7: 49, 8: 64.
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12 for i in range(20, 27, 2): # i takes on the values 20, 22, 24, and 26.
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↓ python3 for_loop_range.py ↓

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- Wir benutzen jeweils `i` als Schleifenvariable.

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1 {0: 0, 1: 1, 2: 4, 3: 9, 4: 16, 6: 36, 7: 49, 8: 64, 20: 400, 22: 484,
  ↪ 24: 576, 26: 676, 40: 1600, 37: 1369, 34: 1156, 31: 961}
2 Hello World!
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4 Hello World!
```

Beispiel (1)



- Schauen wir uns mal ein Beispiel an.
- Zuerst wollen wir ein Dictionary bauen, in dem einige Ganzzahlen ihrem jeweiligen Quadrat zugeordnet sind.
- Wir benutzen vier `for`-Schleifen, um dieses Dictionary mit Daten zu füllen.
- Wir benutzen jeweils `i` als Schleifenvariable.
- In der ersten Schleife iteriert `i` über `range(5)`.

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6  for i in range(5): # i takes on the values 0, 1, 2, 3, 4 -- one by one.
7      squares[i] = i * i # Stores 0: 0, 1: 1, 2: 4, 3: 9, 4: 16.
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9  for i in range(6, 9): # i takes on the values 6, 7, and 8 one by one.
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12     for i in range(20, 27, 2): # i takes on the values 20, 22, 24, and 26.
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15     for i in range(40, 30, -3): # i takes on the values 40, 37, 34, and 31.
16         squares[i] = i * i # Stores 40: 1600, 37: 1369, 34: 1156, 31: 961.
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18     print(squares) # Print the dictionary.
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- Wir benutzen jeweils `i` als Schleifenvariable.
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- Bei der ersten Ausführung des Schleifenkörpers hat `i` den Wert `0`.
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- Der Schleifenkörper `squares[i] = i * i` macht also `squares[1] = 1`.

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Beispiel (1)



- Der Schleifenkörper

`squares[i] = i * i` ist also äquivalent zu `squares[0] = 0` und weist daher den Wert 0 dem Schlüssel 0 im Dictionary `squares` zu.

- In der zweiten Iteration hat `i` den Wert 1.

- Der Schleifenkörper

`squares[i] = i * i` macht also `squares[1] = 1`.

- In der dritten Iteration hat `i` den Wert 2 und wir führen `squares[2] = 4` aus.

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- In der zweiten Iteration hat `i` den Wert `1`.
- Der Schleifenkörper `squares[i] = i * i` macht also `squares[1] = 1`.
- In der dritten Iteration hat `i` den Wert `2` und wir führen `squares[2] = 4` aus.
- Danach gilt `i = 3` und wir machen `squares[3] = 9`.

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- Zuletzt erfolgt dann
`squares[4] = 16` und die erste Schleife ist fertig.

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- Mit der zweiten Schleife iterieren wir über `range(6, 9)`, also es gilt `i = 6`, dann `i = 7`, und schließlich `i = 8`.

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- Mit der zweiten Schleife iterieren wir über `range(6, 9)`, also es gilt `i = 6`, dann `i = 7`, und schließlich `i = 8`.

- Es erfolgt also `squares[6] = 36`, `squares[7] = 49`, und `squares[8] = 64`.

- Die dritte Schleife iteriert über `range(20, 27, 2)` und ihr Körper führt daher nacheinander `squares[20] = 400`, `squares[22] = 484`, `squares[24] = 576`, und `squares[26] = 676` aus.

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führt daher nacheinander
`squares[20] = 400`,
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`squares[26] = 676` aus.
- In der vierten Schleife, nimmt `i` die
Werte der Sequenz
`range(40, 30, -3)` an, die eine
negative Schrittweite hat.

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Werte der Sequenz
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dann `34`, und schließlich `31`.

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9  for i in range(6, 9): # i takes on the values 6, 7, and 8 one by one.
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Beispiel (1)



- Die dritte Schleife iteriert über `range(20, 27, 2)` und ihr Körper führt daher nacheinander

```
squares[20] = 400,  
squares[22] = 484,  
squares[24] = 576, und  
squares[26] = 676 aus.
```

- In der vierten Schleife, nimmt `i` die Werte der Sequenz

`range(40, 30, -3)` an, die eine negative Schrittweite hat.

- `i` wird daher zuerst `40`, dann `37`, dann `34`, und schließlich `31`.

- Nun drucken wir das Dictionary aus und bekommen das erwartete Ergebnis.

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Gute Praxis

Wenn der Wert einer Variable oder eines Parameters egal ist, dann sollten wir diese `_` nennen⁴⁰.

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- Was wir in diesem Programm wieder und wieder tun, ist...

```
1 from math import pi, sqrt    # We need sqrt. pi is for comparison.
2
3 # We use f-strings with Unicode escapes to print the current result.
4 # "\u03c0" is the Unicode escape for the Greek letter pi.
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21 print(f"{e} edges, side length={s} give \u03c0\u2248{e * s / 2}.")
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↓ python3 pi_liu_hui.py ↓

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- Wenn wir all das einfach in eine Schleife packen...

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↓ python3 for_loop_pi_liu_hui.py ↓

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- Und die Ausgabe ist die gleiche!

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continue und break



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- Für den ersten Fall gibt es das Schlüsselwort `continue`, für den zweiten Fall das Schlüsselwort `break`⁴⁹.

Beispiel

- Schauen wir uns mal ein Beispiel an.



Beispiel



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- Wir iterieren mit einer Variable `i` über die 15 Werte von 0 bis 14, also über `range(15)`.

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1 """Here we explore the `break` and `continue` statements."""
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3 for i in range(15): # i takes on the values from 0 to 14 one by one.
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↓ `python3 for_loop_continue_break.py` ↓

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- Normalerweise würden wir dafür die `range` ändern, wir wollen hier aber einfach mal der `break`-Statement verwenden.

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8 All done.
```

Beispiel



- Der letzte Befehl in der Schleife, `print(s)`, gibt diesen String aus.
- Anstatt das wir `i` von 0 bis 14 laufen lassen, wir überlegen uns, dass wir abbrechen wollen, sobald `i` größer als 10 wird.
- Normalerweise würden wir dafür die `range` ändern, wir wollen hier aber einfach mal der `break`-Statement verwenden.
- Wir schreiben es also in eine Alternative hinein, mit Bedingung `if i > 10:`.

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↓ `python3 for_loop_continue_break.py` ↓

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Beispiel



- Also wenn `i > 10`, dann wird `break` ausgeführt, was die Schleife dann sofort beendet.
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- Die Schleife setzt $i = 6$ und das selbe passiert wieder.

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- Wenn `5 <= i <= 8` zutrifft, dann machen wir `continue`.
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- Die Schleife setzt `i = 6` und das selbe passiert wieder.
- Das geht so weiter, bis `i == 9`.
- Die Bedingung `5 <= i <= 8` trifft nämlich **nicht** zu für alle `i ∈ 0..4 ∪ 9..15`.

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- Die Bedingung `5 <= i <= 8` trifft nämlich **nicht** zu für alle `i ∈ 0..4 ∪ 9..15`.
- Das `print(s)` kann also nur in diesen Fällen erreicht werden.

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- Die Schleife setzt `i = 6` und das selbe passiert wieder.
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- Das `print(s)` kann also nur in diesen Fällen erreicht werden.
- Natürlich wird die Schleife abbrechen, sobald `i == 11`.

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- Das geht so weiter, bis `i == 9`.
- Die Bedingung `5 <= i <= 8` trifft nämlich **nicht** zu für alle `i ∈ 0..4 ∪ 9..15`.
- Das `print(s)` kann also nur in diesen Fällen erreicht werden.
- Natürlich wird die Schleife abbrechen, sobald `i == 11`.
- Das Programm wird also `s` nur für `i ∈ 0..4 ∪ {9, 10}` ausgeben, bevor letztendlich `All done.` ausgeben wird.

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- Das `print(s)` kann also nur in diesen Fällen erreicht werden.
- Natürlich wird die Schleife abbrechen, sobald `i == 11`.
- Das Programm wird also `s` nur für $i \in 0..4 \cup \{9, 10\}$ ausgeben, bevor letztendlich `All done.` ausgegeben wird.
- Mit `break` und `continue` haben wir nun zwei neue Werkzeuge, mit denen wir entweder die ganze Schleife abbrechen können oder die aktuelle Iteration beenden (und mit der nächsten weitermachen können, wenn es eine gibt).

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- Probieren wir das mal aus, indem wir eine Liste aller Primzahlen kleiner als 200 erstellen!

Definition: Primzahl

Eine Primzahl (EN: *prime number*) $p \in \mathbb{N}_1$ ist eine positive Ganzzahl $p > 1$, also größer als eins, die keine positiven ganzzahligen Teiler anders als 1 und p selbst hat^{17,61,78}.



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Beispiel



- In unserem Programm wollen wir alle Primzahlen $p < 200$ in einer Liste `primes` speichern.

```
1  """Compute all primes less than 200 using two nested for loops."""
2
3  from math import isqrt # the integer square root == int(sqrt(...))
4
5  primes: list[int] = [2] # the list for the primes; We know 2 is prime.
6  n_divisions: int = 0 # We want to know how many divisions we needed.
7
8  for candidate in range(3, 200, 2): # ...all odd numbers less than 200.
9      is_prime: bool = True # Let us assume that `candidate` is prime.
10
11     for check in range(3, isqrt(candidate) + 1, 2): # ...odd numbers
12         n_divisions += 1 # Every test requires one modulo division.
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Beispiel



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```

Beispiel



- Wenn uns das gelingt, dann setzen wir `is_prime = False`.
- Wenn wir keinen Teiler finden, dann bleibt `is_prime True`.
- Dann würden wir `candidate` zur Liste `primes` hinzufügen.
- Das ist der Plan.

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1  """Compute all primes less than 200 using two nested for loops."""
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3  from math import isqrt # the integer square root == int(sqrt(...))
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Beispiel



- Wenn wir keinen Teiler finden, dann bleibt `is_prime` `True`.
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- Das ist der Plan.
- Wir werden diesen Plan mit einer geschachtelten / inneren Schleife implementieren.

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Beispiel



- Dann würden wir `candidate` zur Liste `primes` hinzufügen.
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- Wir werden diesen Plan mit einer geschachtelten / inneren Schleife implementieren.
- Weil die Schleifenvariable `candidate` immer ungerade ist, kommen auch nur ungerade Zahlen als Teiler in Frage.

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- In Python kann so eine „abgerundete“ Quadratwurzel $\lfloor \sqrt{a} \rfloor$ einer Ganzzahl a mit der Funktion `isqrt` aus dem Modul `math` berechnet werden.

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Beispiel



- Die meisten Ganzzahlen haben keine ganzzahligen Quadratwurzeln.
- Da ganzzahlige Teiler aber keine Nachkommastellen haben können, reicht es aus, wenn wir $\lfloor \sqrt{\text{candidate}} \rfloor$ als Obergrenze nehmen.
- In Python kann so eine „abgerundete“ Quadratwurzel $\lfloor \sqrt{a} \rfloor$ einer Ganzzahl a mit der Funktion `isqrt` aus dem Modul `math` berechnet werden.
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Beispiel



- Dann können `is_prime = False` setzen.
- Genaugenommen können wir dann auch gleich die innere Schleife mit `break` abbrechen.
- Sobald wir wissen, dass `candidate` keine Primzahl ist, müssen wir keine weiteren potentiellen Teiler prüfen.
- Beachten Sie, dass wir alle Modulo-Divisionen zählen, in dem wir `n_divisions += 1` am Anfang der inneren Schleife machen.

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5  primes: list[int] = [2] # the list for the primes; We know 2 is prime.
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8  for candidate in range(3, 200, 2): # ...all odd numbers less than 200.
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↓ python3 for_loop_nested_primes.py ↓

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- Wir erfahren, dass wir mit 252 Divisionen alle 46 Primzahlen im Intervall 2..199 finden konnten.
- (OK, wir haben ignoriert bzw. wissen nicht, ob `isqrt` irgendwelche Divisionen durchführt, aber egal.)

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- Wir haben auch gesagt, dass wir schon verschiedene Kontainerdatentypen kennen, die wir als Sequenzen behandeln können.
- Wir müssten also über `lists`, `tuples`, `sets`, und `dicts` iterieren können...

Beispiel (1)

- Schauen wir uns das mal an.



Beispiel (1)



- Schauen wir uns das mal an.
- In unserem Beispielprogramm bauen wir uns eine Liste `txt` mit Strings zusammen, die wir später ausgeben wollen.

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1  """Iterate over several different containers with `for` loops."""
2
3  txt: list[str] = [] # We will collect the output text in this list.
4
5  lst: list[int] = [1, 2, 3, 50] # Create a list with 4 integers.
6  for i in lst: # i takes on the values 1, 2, 3, and 50.
7      txt.append(f"{i} = ") # We store "i = 1", "i = 2", "i = 3"...
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9  tp: tuple[float, ...] = (7.6, 9.4, 8.1) # Create a tuple with 3 floats.
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25 # Merge text into single string with separator ", " and print it.
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↓ python3 for_loop_sequence.py ↓

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1 i = 1, i = 2, i = 3, i = 50, f = 7.6, f = 9.4, f = 8.1, s = 'u', s = 'u'
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- Schauen wir uns das mal an.
- In unserem Beispielprogramm bauen wir uns eine Liste `txt` mit Strings zusammen, die wir später ausgeben wollen.
- Zuerst iterieren wir über eine Liste `lst` mit den vier Zahlen `[1, 2, 3, 50]`.

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- Zuerst iterieren wir über eine Liste `lst` mit den vier Zahlen `[1, 2, 3, 50]`.
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- Im Körper (EN: *body*) der Schleife rufen wir dann `txt.append(f"{i = }")` auf.

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Beispiel (1)



- Das geht mit `for i in lst...`
- `i` ist die Schleifenvariable und wird Schritt-für-Schritt alle Werte aus `lst` annehmen.
- Im Körper (EN: *body*) der Schleife rufen wir dann `txt.append(f"{i = }")` auf.
- Dieser f-String ergibt `"i = 1"` in der ersten Iteration, `"i = 2"` in der zweiten, `"i = 3"` in der dritten, und `"i = 50"` in der letzten Iteration.

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3  txt: list[str] = [] # We will collect the output text in this list.
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6  for i in lst: # i takes on the values 1, 2, 3, and 50.
7      txt.append(f"{i = }") # We store "i = 1", "i = 2", "i = 3"...
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↓ python3 for_loop_sequence.py ↓

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Beispiel (1)



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Beispiel (1)



- Hier funktioniert es ganz genauso.
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- Erinnern Sie sich: Mengen sind in Python ungeordnet.

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↓ python3 for_loop_sequence.py ↓

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Beispiel (1)



- Wir können über die Elemente dieser Menge iterieren, in dem wir `for s in st` schreiben.
- Die Schleifenvariable `s` nimmt dann die Werte `"w"`, `"u"`, und `"v"` in beliebiger Reihenfolge an.
- Erinnern Sie sich: Mengen sind in Python ungeordnet.
- Wenn wir das Programm zweimal ausführen, können wir also eventuell verschiedene Reihenfolgen beobachten.

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↓ python3 for_loop_sequence.py ↓

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   ↳ False
```


Beispiel (1)



- Es beinhaltet nur die beiden Einträge `{1.1: True, 2.5: False}`.
- Dictionaries sind etwas speziell.
- Sie ordnen Werte zu Schlüsseln zu.
- Wenn wir mit dem ganzen Dictionary `dc` als Kollektion arbeiten, dann können wir auf drei Arten darauf zugreifen.

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- Dictionaries sind etwas speziell.
- Sie ordnen Werte zu Schlüsseln zu.
- Wenn wir mit dem ganzen Dictionary `dc` als Kollektion arbeiten, dann können wir auf drei Arten darauf zugreifen.
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Beispiel (1)



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Beispiel (1)



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Beispiel (1)



- Wir iterieren über die Werte mit `for v in dc.values()`, wodurch `v` erst `True` und dann `False` wird.
- Zu guter Letzt iterieren wir über die Schlüssel-Wert-Paare.
- Schauen Sie genau hin!
- Wir *könnten* schreiben `for t in dc.items()`, wodurch wir eine Variable `t` die Werte `(1.1, True)` und dann `2.5: False` annehmen würde.

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↓ python3 for_loop_sequence.py ↓

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- Wir *könnten* schreiben `for t in dc.items()`, wodurch wir eine Variable `t` die Werte `(1.1, True)` und dann `2.5: False` annehmen würde.
- Aber wir haben ja vom automatischen „Auspacken“ von Tupeln gelernt.

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- Und wieder hängen wir diese als Text an unsere Liste `txt` an.

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- Nach diesen ganzen Schleifen haben wir nun eine Liste `txt` mit 16 Strings.

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- Das könnten wir mit einer Schleife machen.
- Python bietet aber eine einfachere und schnellere Methode dafür an.

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6  for i in lst: # i takes on the values 1, 2, 3, and 50.
7      txt.append(f"{i} = ") # We store "i = 1", "i = 2", "i = 3"...
8
9  tp: tuple[float, ...] = (7.6, 9.4, 8.1) # Create a tuple with 3 floats.
10 for f in tp: # f takes on the values 7.6, 9.4, and 8.1.
11     txt.append(f"{f} = ") # We store "f = 7.6", "f = 9.4", ...
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13 st: set[str] = {"u", "v", "w"} # Create a set with 3 strings.
14 for s in st: # s takes on the values "u", "v", and "w" (unordered!).
15     txt.append(f"s = {s!r}") # We store "s = 'u'", "s = 'v'", ...
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↓ python3 for_loop_sequence.py ↓

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Beispiel (1)



- Wir wollen diese zu einzigen Zeichenkette zusammenführen, wobei wir `"", "` als Separator benutzen wollen.
- Das könnten wir mit einer Schleife machen.
- Python bietet aber eine einfachere und schnellere Methode dafür an.
- Die Methode `join` der Klasse `str`.

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Beispiel (1)



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- Daher produziert `", ".join(txt)` als Ergebnis
`i = 1, i = 2, i = 3, i = 50,`
`f = 7.6, ....`
- Dieser Text wird mit der `print` ausgegeben.

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Beispiel (2)



- Verwenden wir das Konzept der Iteration über Sequenzen nun für etwas Sinnvolles.

```
1  """Compute all primes less than 200 using two nested for loops."""
2
3  from math import isqrt # the integer square root == int(sqrt(...))
4
5  primes: list[int] = [2] # the list for the primes; We know 2 is prime.
6  n_divisions: int = 0 # We want to know how many divisions we needed.
7
8  for candidate in range(3, 200, 2): # ...all odd numbers less than 200.
9      is_prime: bool = True # Let us assume that `candidate` is prime.
10
11     for check in range(3, isqrt(candidate) + 1, 2): # ...odd numbers
12         n_divisions += 1 # Every test requires one modulo division.
13         if candidate % check == 0: # modulo == 0: division without rest
14             is_prime = False # check divides candidate evenly, so
15             break # candidate is not prime. We can stop the inner loop.
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17     if is_prime: # If True: no smaller number divides candidate evenly.
18         primes.append(candidate) # Store candidate in primes list.
19
20 # Finally, print the list of prime numbers.
21 print(f"After {n_divisions} divisions: {len(primes)} primes {primes}.")
```

↓ python3 for_loop_nested_primes.py ↓

```
1 After 252 divisions: 46 primes [2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31,
  ↪ 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97, 101, 103,
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Beispiel (2)



- Verwenden wir das Konzept der Iteration über Sequenzen nun für etwas Sinnvolles.
- Wir verbessern unser Programm zum Finden von Primzahlen.

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- Als wir mit dem Originalprogramm angefangen haben, dann haben wir sofort die Zahl 2 als eine Primzahl in unsere Liste `primes` gespeichert.

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- Für jede potentielle Primzahl `candidate`, haben wir alle (ungeraden) Zahlen `check` aus `range(3, isqrt(candidate)+ 1, 2)` als mögliche Divisoren getestet.

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- Wenn wir darüber nachdenken, erkennen wir, dass wir eigentlich nur Primzahlen als mögliche Divisoren testen müssen.

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- Wenn wir darüber nachdenken, erkennen wir, dass wir eigentlich nur Primzahlen als mögliche Divisoren testen müssen.
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Beispiel (2)

- Die äußere Schleife wird Schritt-für-Schritt Primzahlen an `primes` anhängen.
- Für jeden Wert der Schleifenvariablen `candidate`, beinhaltet `primes` alle Primzahlen, die kleiner als `candidate` sind (außer 2).
- Natürlich müssen wir nur die Werte von `check` prüfen, die kleiner oder gleich `isqrt(candidate)` sind.
- Darum speichern wir diesen Wert in einer neuen Variablen `limit`, so dass wir ihn nicht in der inneren Schleife wieder und wieder berechnen müssen.

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5  primes: list[int] = [] # The list for the primes is initially empty.
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8  for candidate in range(3, 200, 2): # ...all odd numbers less than 200.
9      is_prime: bool = True # Let us assume that `candidate` is prime.
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12     for check in primes: # We only test with the odd primes we got.
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- Mit `break` können wir dann die innere Schleife abbrechen, wenn wir dieses `limit` erreichen.
- Nach dem die äußere Schleife fertig ist, fügen wir noch die 2 an Index 0 in die Liste ein.
- Die Liste ist dann genau die selbe, wie beim Originalprogramm.
- Allerdings brauchen wir nun nur 224 Divisionen anstatt von 252.

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- Wir können auch mit `continue` in die nächste Iteration springen.
- Und wir können Schleifen und Alternativen beliebig ineinander verschachteln.



谢谢你们！
Thank you!
Vielen Dank!



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Glossary (in English) I



Bash is a shell used under Ubuntu Linux, i.e., the program that „runs“ in the terminal and interprets your commands, allowing you to start and interact with other programs^{11,50,82}. Learn more at <https://www.gnu.org/software/bash>.

C is a programming language, which is very successful in system programming situations^{22,56,76}.

client In a client-server architecture, the client is a device or process that requests a service from the server. It initiates the communication with the server, sends a request, and receives the response with the result of the request. Typical examples for clients are web browsers in the internet as well as clients for database management systems (DBMSes), such as `psql`.

client-server architecture is a system design where a central server receives requests from one or multiple clients^{6,43,53,57,60}. These requests and responses are usually sent over network connections. A typical example for such a system is the World Wide Web (WWW), where web servers host websites and make them available to web browsers, the clients. Another typical example is the structure of database (DB) software, where a central server, the DBMS, offers access to the DB to the different clients. Here, the client can be some terminal software shipping with the DBMS, such as `psql`, or the different applications that access the DBs.

DB A *database* is an organized collection of structured information or data, typically stored electronically in a computer system. Databases are discussed in our book *Databases*⁷⁵.

DBMS A *database management system* is the software layer located between the user or application and the DB. The DBMS allows the user/application to create, read, write, update, delete, and otherwise manipulate the data in the DB⁸¹.

escape sequence Escaping is the process of presenting „forbidden“ characters or symbols in a sequence of characters or symbols. In Python⁷⁷, string escapes allow us to include otherwise impossible characters, such as string delimiters, in a string. Each such character is represented by an *escape sequence*, which usually starts with the backslash character („\“)²⁴. In Python strings, the escape sequence `\"`, for example, stands for `"`, the escape sequence `\\` stands for `\`, and the escape sequence `\n` stands for a newline or linebreak character. In Python f-strings, the escape sequence `{ }` stands for a single curly brace `{`. In PostgreSQL⁷⁵, similar C-style escapes (starting with „\“) are supported⁶⁹.

Glossary (in English) II



f-string let you include the results of expressions in strings^{12,29–31,47,64}. They can contain expressions (in curly braces) like `f"a{6-1}b"` that are then transformed to text via (string) interpolation, which turns the string to `"a5b"`. F-strings are delimited by `f"..."`.

IT information technology

LAMP Stack A system setup for web applications: Linux, Apache (a web server), MySQL, and the server-side scripting language PHP^{13,36}.

Linux is the leading open source operating system, i.e., a free alternative to Microsoft Windows^{3,35,63,73,74}. We recommend using it for this course, for software development, and for research. Learn more at <https://www.linux.org>. Its variant Ubuntu is particularly easy to use and install.

MariaDB An open source relational database management system that has forked off from MySQL^{1,2,4,23,45,58}. See <https://mariadb.org> for more information.

Microsoft Windows is a commercial proprietary operating system¹⁰. It is widely spread, but we recommend using a Linux variant such as Ubuntu for software development and for our course. Learn more at <https://www.microsoft.com/windows>.

modulo division is, in Python, done by the operator `%` that computes the remainder of a division. `15 % 6` gives us 3.

MySQL An open source relational database management system^{9,23,59,71,80}. MySQL is famous for its use in the LAMP Stack. See <https://www.mysql.com> for more information.

PostgreSQL An open source object-relational DBMS^{27,52,55,71}. See <https://postgresql.org> for more information.

psql is the client program used to access the PostgreSQL DBMS server.

Python The Python programming language^{37,42,44,77}, i.e., what you will learn about in our book⁷⁷. Learn more at <https://python.org>.

Glossary (in English) III



relational database A relational DB is a database that organizes data into rows (tuples, records) and columns (attributes), which collectively form tables (relations) where the data points are related to each other^{16,32,34,65,70,75,79}.

server In a client-server architecture, the server is a process that fulfills the requests of the clients. It usually waits for incoming communication carrying the requests from the clients. For each request, it takes the necessary actions, performs the required computations, and then sends a response with the result of the request. Typical examples for servers are web servers¹³ in the internet as well as DBMSes. It is also common to refer to the computer running the server processes as server as well, i.e., to call it the „server computer“⁴¹.

SQL The *Structured Query Language* is basically a programming language for querying and manipulating relational databases^{14,18–20,38,48,66–68,70}. It is understood by many DBMSes. You find the Structured Query Language (SQL) commands supported by PostgreSQL in the reference⁶⁶.

(string) interpolation In Python, string interpolation is the process where all the expressions in an f-string are evaluated and the final string is constructed. An example for string interpolation is turning `f"Rounded {1.234:.2f}"` to `"Rounded 1.23"`.

terminal A terminal is a text-based window where you can enter commands and execute them^{3,15}. Knowing what a terminal is and how to use it is very essential in any programming- or system administration-related task. If you want to open a terminal under Microsoft Windows, you can Druck auf  + , dann Schreiben von `cmd`, dann Druck auf . Under Ubuntu Linux,  +  +  opens a terminal, which then runs a Bash shell inside.

Ubuntu is a variant of the open source operating system Linux^{15,36}. We recommend that you use this operating system to follow this class, for software development, and for research. Learn more at <https://ubuntu.com>. If you are in China, you can download it from <https://mirrors.ustc.edu.cn/ubuntu-releases>.

WWW World Wide Web^{5,21}

Glossary (in English) IV



- π is the ratio of the circumference U of a circle and its diameter d , i.e., $\pi = U/d$. $\pi \in \mathbb{R}$ is an irrational and transcendental number^{28,39,51}, which is approximately $\pi \approx 3.141\,592\,653\,589\,793\,238\,462\,643$. In Python, it is provided by the `math` module as constant `pi` with value `3.141592653589793`⁴⁶. In PostgreSQL, it is provided by the SQL function `pi()` with value `3.141592653589793`⁴⁶.
- $i..j$ with $i, j \in \mathbb{Z}$ and $i \leq j$ is the set that contains all integer numbers in the inclusive range from i to j . For example, $5..9$ is equivalent to $\{5, 6, 7, 8, 9\}$
- e is Euler's number²⁶, the base of the natural logarithm. $e \in \mathbb{R}$ is an irrational and transcendental number^{28,39}, which is approximately $e \approx 2.718\,281\,828\,459\,045\,235\,360$. In Python, it is provided by the `math` module as constant `e` with value `2.718281828459045`. In PostgreSQL, you can obtain it via the SQL function `exp(1)` as value `2.718281828459045`⁴⁶.
- $\mathbb{N}_1$ the set of the natural numbers *excluding* 0, i.e., 1, 2, 3, 4, and so on. It holds that $\mathbb{N}_1 \subset \mathbb{Z}$.
- $\mathbb{R}$ the set of the real numbers.
- $\mathbb{Z}$ the set of the integers numbers including positive and negative numbers and 0, i.e., $\dots, -3, -2, -1, 0, 1, 2, 3, \dots$, and so on. It holds that $\mathbb{Z} \subset \mathbb{R}$.